
Task #1

You leave from home on a road trip at **noon**. You are driving steadily at 25 m/s for **180 km**. After that distance, it begins to rain, so you slow down to 18 m/s for the rest of the trip. At **4:30pm**, you arrive at your destination. How far is your destination from home?

Adapted from problem 2-7 in Giancoli *Physics: Principles and Applications*, 7th edition.

Task #2

Two bicycles start moving, but one has a starting position that is ahead of the other. The front bicycle travels at a constant speed of 12 m/s, while the back bicycle travels at a constant speed of 18 m/s. The initial distance between the two bicycles is 150 meters. Calculate where the back bicycle overtakes the front bicycle.

Task #3

Two cars start a distance of 100 meters apart, facing each other. Car A travels east at 10 m/s; Car B travels west at 15 m/s.

Superfly is sitting on the front bumper of Car A. He has super powers that allow him to fly at a constant speed of 45 m/s and instantaneously change direction. Superfly flies toward the bumper of Car B and then instantaneously turns around and flies back toward Car A. He continues to go back and forth between the two cars until they collide. (Superfly is also invincible.)

What distance does Superfly travel before the cars collide?
